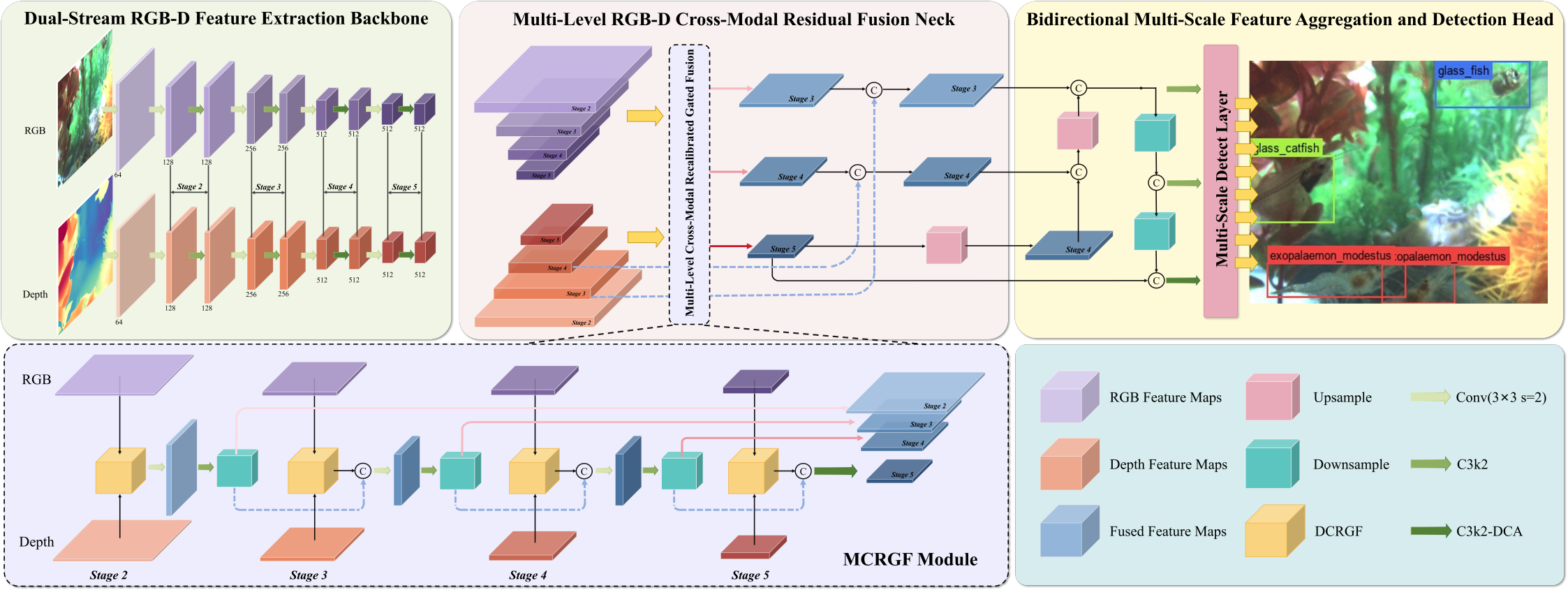

Dual-Stream RGB-D Feature Extraction Backbone
RGB
Depth
64
128
128
256
256
512
512
512
512
Stage 2
Stage 3
Stage 4
Stage 5
Multi-Level RGB-D Cross-Modal Residual Fusion Neck
Multi-Level Cross-Modal Recalibrated Gated Fusion
Stage 2
Stage 3
Stage 4
Stage 5
C
Bidirectional Multi-Scale Feature Aggregation and Detection Head
Multi-Scale Detect Layer
glass_fish
glass_catfish
exopalaemon_modestus
opalaemon_modestus
RGB
Depth
Stage 2
Stage 3
Stage 4
Stage 5
MCRGF Module
RGB Feature Maps
Depth Feature Maps
Fused Feature Maps
Upsample
Downsample
DCRGF
Conv(3×3 s=2)
C3k2
C3k2-DCA